

WEEK 5 – POWER BI EDA REPORT

HEALTHCARE DATA TRANSFORMATION & DATA MODELING

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Department: Data Science and Artificial Intelligence

Tool Used: Microsoft Power BI

Project: Healthcare Data Transformation,& Data Modeling.

1. INTRODUCTION

Power BI is a business intelligence and data visualization tool that helps in transforming, organizing, and analysing data. In this Week 5 task, I worked with the healthcare_dataset.csv dataset using Microsoft Power BI.

I used Power Query to clean and transform the healthcare data and created an Age Group column using a Conditional Column. After transforming the data, I organized the data into separate tables and created a structured data model. I also established Many-to-One (*:1) relationships between the related tables.

This task helped me gain practical knowledge of Power Query, data transformation, Conditional Columns, Data Modeling, and Relationships in Power BI.

2. OBJECTIVE

The main objective of this Week 5 task is to prepare and organize the healthcare dataset using Microsoft Power BI.

- To clean and transform the healthcare data using Power Query.
- To apply different data transformation techniques to improve data quality.
- To create an **Age Group** column using a Conditional Column.
- To organize the healthcare data into meaningful tables using Data Modeling.
- To establish **Many-to-One (*:1) relationships** between the tables.
- To prepare a structured data model for further analysis.

3. DATASET

For this task, I used the healthcare_dataset.csv dataset in Microsoft Power BI.

The dataset contains healthcare-related information such as:

- Patient ID
- Name
- Age
- Gender
- Blood Type
- Medical Condition
- Date of Admission
- Doctor
- Hospital
- Insurance Provider
- Billing Amount
- Room Number
- Admission Type
- Discharge Date
- Medication
- Test Results
- Admission ID

I used this dataset to perform data transformation and data modeling in Power BI.

healthcare_dataset	Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor
Dim_Admission	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
Dim_Patient	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
Billing	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
Measures [2]							
1	Bobby Jackson	30	MALE	B-	Cancer	31-01-2024	Matthew Gray
2	Leslie Terry	60	MALE	A+	Obesity	20-06-2023	Kenneth De
3	Danny Smith	76	FEMALE	A-	Obesity	22-09-2022	Tiffany Mittle
4	Andrew Watts	28	FEMALE	O+	Diabetes	18-11-2020	Kenneth Wells
5	Adrianne Bell	42	FEMALE	AB+	Cancer	19-09-2022	Kathleen Hah
6	Emily Johnson	66	MALE	A+	Asthma	20-12-2023	Taylor Hewitt
7	Edward Edwards	23	FEMALE	AB-	Diabetes	25-11-2020	Kelly Olson
8	Christine Martinez	20	FEMALE	A+	Cancer	28-12-2022	Suzanne Ther
9	Jaimee Aguilar	62	MALE	AB+	Asthma	01-07-2020	Daniel Fergus
10	Christopher Berg	58	FEMALE	AB-	Cancer	29-05-2021	Heather Day
11	Michelle Daniels	72	MALE	O+	Cancer	19-04-2020	John Duncan
12	Karen Martinez	38	FEMALE	A-	Hypertension	19-06-2021	Douglas Mon
13	Connor Hansen	75	FEMALE	A+	Diabetes	12-12-2023	Kenneth Hah
14	Robert Bauer	68	FEMALE	AB+	Asthma	22-03-2020	Theresa Flier
15	Brooke Brady	44	FEMALE	AB+	Cancer	08-10-2021	Robert Steu
16	Mrs. Malika Daniels	48	FEMALE	AB-	Obesity	01-01-2023	Maria Dought
17	Wiley Perkins	62	FEMALE	A+	Arthritis	23-06-2020	Erica Spencer
18	Mrs. Jenna Campbell	38	MALE	AB-	Obesity	08-04-2020	Jenna Kim
19	Luke Burgess	34	FEMALE	A-	Hypertension	04-05-2021	Lydia Moore
20	Daniel Schmidt	69	MALE	B+	Asthma	18-11-2022	Dennis Gubka
21	Timothy Barnes	67	FEMALE	A-	Asthma	28-06-2023	Krista Smith
22	Christopher Bright	48	MALE	B+	Asthma	21-01-2020	Gregory Smith
23	Bartley Stewart	52	FEMALE	O+	Arthritis	12-05-2022	Vanessa New
24	Dr. Ellen Thompson	89	MALE	A+	Asthma	02-06-2021	Doreen Martin
25	Paul Henderson	72	FEMALE	AB+	Hypertension	25-05-2020	Stephania Kro
26		71	MALE	A+			

4. POWER QUERY DATA TRANSFORMATION & AGE GROUP

I used Power Query Editor to clean and transform the healthcare dataset before creating the data model.

4.1 Data Transformation Steps

The following transformation steps were performed:

1. Promoted Headers

I promoted the first row as the column headers so that each field could be identified correctly.

2. Changed Type

I changed the data types of the columns according to their values, such as text, number, and date.

3. Capitalized Each Word

I formatted the required text values by capitalizing each word to maintain consistent formatting.

4. Removed Errors

I removed error values from the dataset to improve data quality.

5. Trimmed Text

I removed unnecessary spaces from text values using the Trim transformation.

6. Cleaned Text

I cleaned unwanted or non-printable characters from the text data.

7. Uppercased Text

I converted the required text values into uppercase for consistent data formatting.

8. Added Index

I added an Index Column to provide sequential index values and support the organization of the data.

9. Removed Columns

I removed unnecessary columns that were not required for the task.

10. Renamed Columns

I renamed the required columns to make the field names clear and meaningful.

11. Removed Duplicates

I removed duplicate records to improve the consistency and quality of the dataset.

	Name	Gender	Blood Type	National Origin	Date of Birth
1	Billy Nelson	MALE	B-	Swedish	31-07-2001, Matthew Lee
2	Linda Terry	FEMALE	A+	Swedish	25-08-2001, Samantha G
3	Steve Smith	MALE	A-	Swedish	22-05-2001, Tiffany Brown
4	Adrian Wells	MALE	O+	Swedish	28-11-2001, Kevin Wells
5	Andrew Bell	MALE	AB+	Swedish	29-08-2001, Nathan Ross
6	John Wilson	MALE	A+	Swedish	20-12-2001, Taylor Smith
7	Edward Edwards	MALE	AB-	Swedish	05-11-2001, Kelly Stone
8	Christina Matthews	FEMALE	A+	Swedish	08-10-2001, Nicholas Thor
9	Samuel Appleton	MALE	AB-	Swedish	01-01-2001, Daniel Fergus
10	Christopher King	MALE	AB-	Swedish	29-05-2001, Matthew Day
11	Michelle Barnes	MALE	O+	Swedish	09-04-2001, John Carlson
12	Adam Mitchell	FEMALE	A-	Hyperborean	07-08-2001, Douglas May
13	Connor Hansen	MALE	A+	Swedish	13-12-2001, Kenneth Hall
14	Robert Brown	MALE	AB+	Swedish	02-01-2001, Theresa Price
15	Wendy Brady	FEMALE	AB+	Swedish	08-10-2001, Robert Stone
16	Mr. Nathan Graham	MALE	AB-	Swedish	01-10-2001, Maria Douglas
17	Julia Peters	FEMALE	A+	Austrian	27-04-2001, Erik Jensen
18	Mr. Isaac Campbell	MALE	AB-	Swedish	06-03-2001, John Kim
19	John Rogers	MALE	A-	Hyperborean	06-03-2001, Justin Moore
20	Samuel Schwartz	MALE	B+	Astorian	23-11-2001, Simon Galtier
21	Timothy Adams	FEMALE	A-	Astorian	28-06-2001, Kyla Smith
22	Christopher Smith	MALE	B+	Astorian	21-01-2001, Gregory Smith
23	Kathryn Stewart	FEMALE	O+	Astorian	22-09-2001, Vanessa Meier
24	Dr. Allen Thompson	MALE	A+	Astorian	02-08-2001, Norma Martin
25	Paul Henderson	FEMALE	AB+	Hyperborean	15-05-2001, Stephanie Gray
26	Mr. William Lee	MALE	O+	Hyperborean	05-04-2001, Erik Jensen

4.2 Age Group Using Conditional Column

I created a new column named Age Group using the Conditional Column feature in Power Query.

The Age column was used to classify patients into different age groups.

Age Condition Age Group

Age $\leq$ 12 Child

Age $\leq$ 19 Teenager

Age $\leq$ 39 Younger Adult

Age $\leq$ 59 Elder Adult

Age $>$ 59 Senior Adult

This classification makes it easier to group patients based on their age.

Add Conditional Column

Add a conditional column that is computed from the other columns or values:

New column name:

Age Group

	Column Name	Operator	Value		Output
If	Age	is less than or equ...	ABC 123 = 12	Then	ABC 123 = child
Else If	Age	is less than or equ...	ABC 123 = 19	Then	ABC 123 = Teenager
Else If	Age	is less than or equ...	ABC 123 = 39	Then	ABC 123 = Younger Adult
Else If	Age	is less than or equ...	ABC 123 = 59	Then	ABC 123 = Elder Adult

Add Clause

Else If

ABC 123 = Senior Adult

OK

Cancel

5. DATA MODELING

After completing the data transformation, I created a structured data model in Power BI.

I organized the healthcare data into the following tables:

Dim_Patient

This table contains patient-related information such as:

- Patient ID
- Name
- Age
- Age Group
- Gender
- Blood Type

Dim_Admission

This table contains admission and medical-related information such as:

- Admission ID
- Admission Type
- Doctor
- Hospital
- Medical Condition
- Medication
- Room Number
- Test Results

Billing

This table contains billing and admission-related information such as:

- Patient ID
- Admission ID
- Billing Amount
- Date of Admission
- Discharge Date
- Insurance Provider

Creating separate tables helped me organize the healthcare data into a structured model and made the data easier to manage for further analysis.



6. RELATIONSHIPS

After creating the tables, I established relationships between the tables using their common key fields.

Relationship 1 – Patient

Billing → Dim_Patient

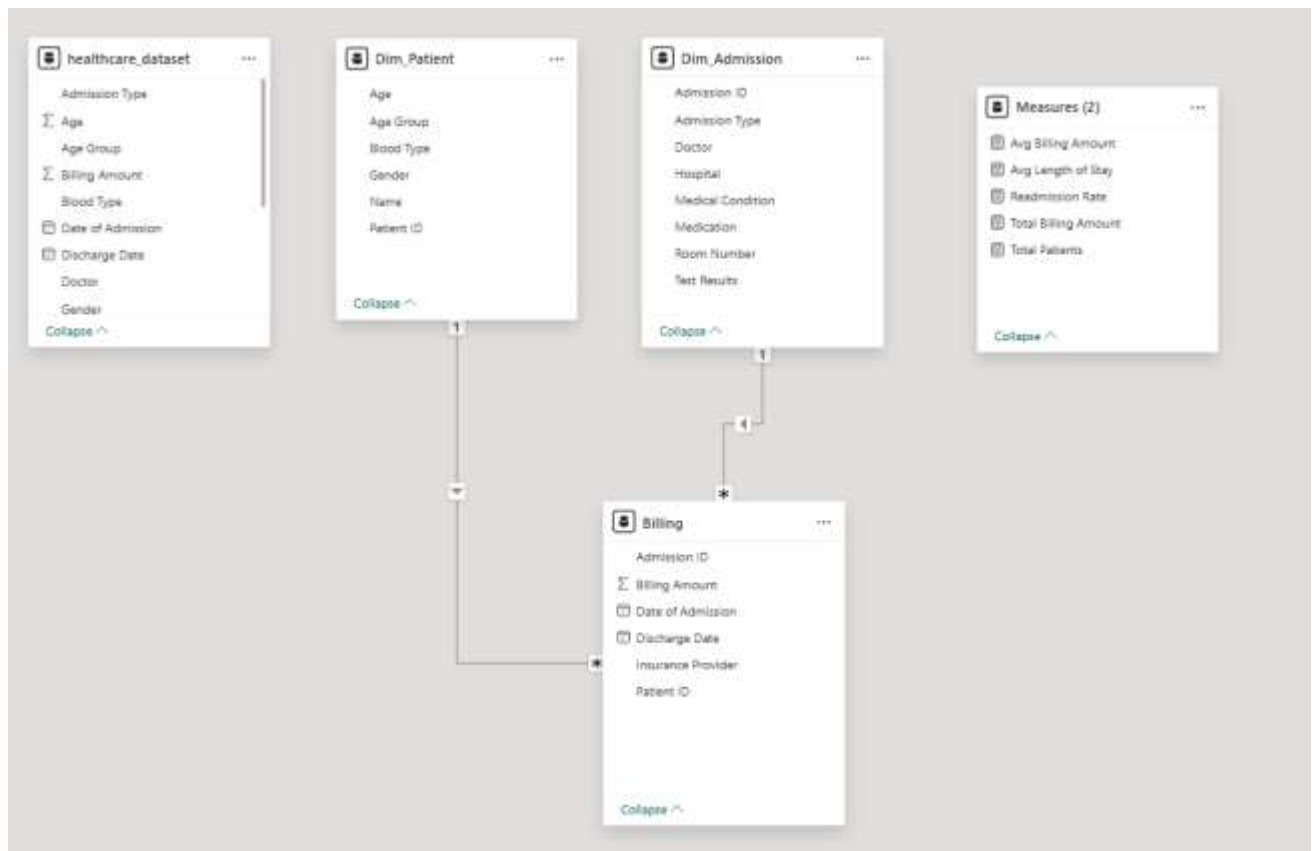
- Common Key: Patient ID
- Cardinality: Many-to-One (*:1)

Relationship 2 – Admission

Billing → Dim_Admission

- Common Key: Admission ID
- Cardinality: Many-to-One (*:1)

The relationships connect the tables and allow related information to be used together during analysis.



7. CONCLUSION

In Week 5, I cleaned and transformed the healthcare dataset using Power Query and created an Age Group using a Conditional Column. I then organized the data into Dim_Patient, Dim_Admission, and Billing tables and established Many-to-One (*:1) relationships.

This task helped me gain practical knowledge of Data Transformation, Data Modeling, and Relationships in Power BI.

